



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

Human Resource Analytics

Lecture 03

Data Science Framework

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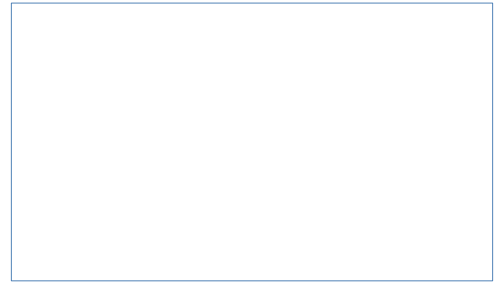
XLRI-Jamshedpur



Content

- Define the goal (HR problem or Issue)
- Definition of HR problem should cover
- Collect and manage data
- Build the model
- Evaluate and critique the model
- Present results
- Deploy model

Data Science Framework

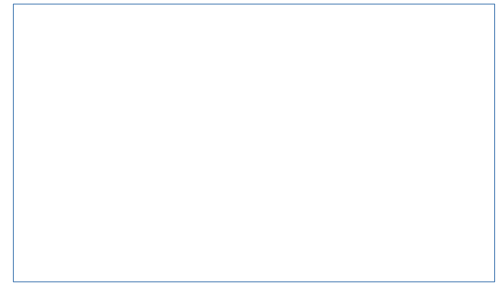


~~Define the goal (HR problem or Issue)~~

Being very clear on this is very important because the specific choices we make further on about data and tools is fundamentally dependent on this.

If our questions are ill-conceived or not clearly delineated, we may lose lot of time in subsequent steps and achieve no value from this effort.

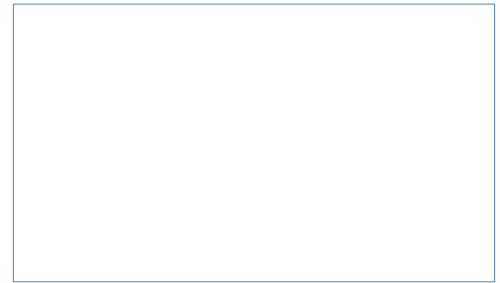
Addition
Selection



Definition of HR problem should cover

- A statement of what ✓
- A statement of why ✓
- A statement of a desired state or outcome we are trying to achieve ✓
- **Example - Job classification / HR operations efficiency** ✓

metrics



Collect and manage data

HRMIS may be good source of HR data in any organization

In HRMIS data may not be in exact form in which we want it

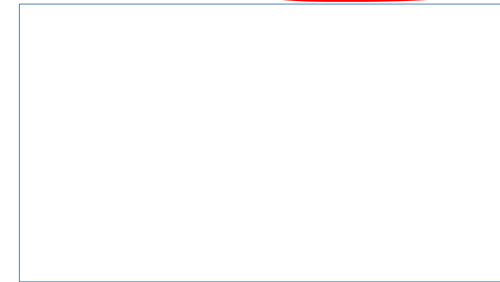
Analytics and statics required to structure our data into the metric (dependent variable or label) and the independent variables (features)

To prepare our data set before the analysis, we have to extract,

transform and load (ETL) Ensure data sources

should be trustworthy otherwise (garbage in and garbage out)

work exp / Dis
- Salary
Crt, Salary, Locn

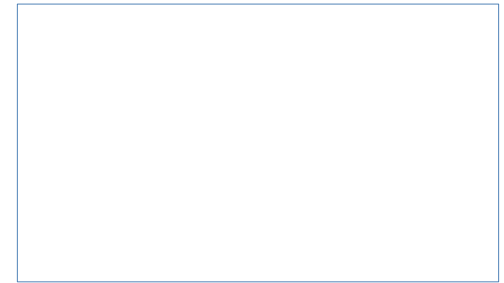


- **Build the model** → metrics

✓ Input – our data ✓

✓ Processing step – Analytical tool – Descriptive, diagnostic, predictive and prescriptive

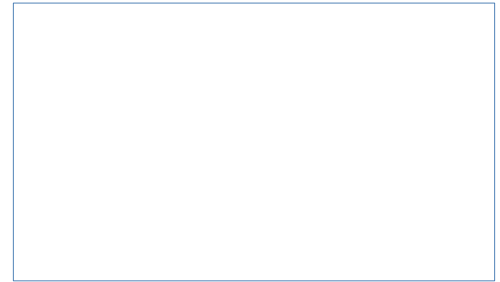
✓ Output – Statistical result ✓



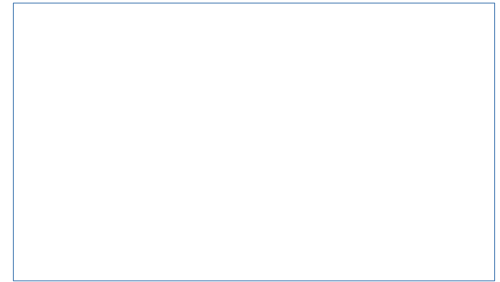
- Evaluate and critique the model

No. of
Application
1000

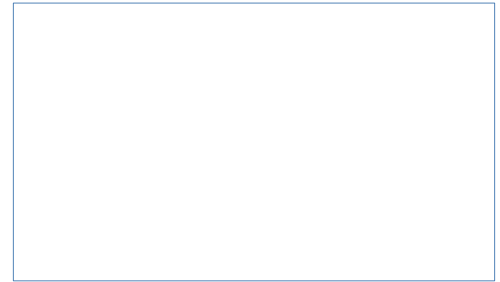
Quality of Hire



- **Present results**



- **Deploy model**



Thank You

